Arindom Aich

Ph.D. Scholar (CSE), IIT Kharagpur

✉ arindomaich2@gmail.com | in LinkedIn | ⊕ Portfolio

PROFESSIONAL SUMMARY

Ph.D. scholar in Computer Science and Engineering and Project Research Assistant at IIT Kharagpur, with 2+ years of industry experience in Generative AI and backend development (Java, Spring Boot, Python, FastAPI, LLM and RAG pipelines). Delivered products across Fashion Tech, Real Estate, and Personal Wellness, and led cross-functional teams from development through deployment. Research interests: Artificial Intelligence, IoT, and Computer Networks.

EDUCATION

Indian Institute of Technology (IIT) Kharagpur — Kharagpur, India
Doctor of Philosophy (Ph.D.), Computer Science and Engineering — *October 2026 – Present*

Tezpur University — Tezpur, Assam
Bachelor of Technology, Computer Science and Engineering | CGPA: 8.04 — *November 2020 – June 2024*

Ramanujan Junior College — Nagaon, Assam
Higher Secondary (AHSEC) | 79.20% — *June 2017 – June 2019*

Sankardev Shishu/Vidya Niketan — Biswanath Chariali, Assam
HSLC (SEBA) | 88.83% — *January 2005 – May 2017*

PROFESSIONAL EXPERIENCE

Project Research Assistant — Kharagpur, India
CoE for Unified AI-Enabled Craft Ecosystem Platform, IIT Kharagpur — *June 2026 – Present*

- Conducting applied research and development on an AI-enabled craft ecosystem platform as part of a Centre of Excellence at IIT Kharagpur.

Software Development Engineer – 1 — Bengaluru, India
Octopi Labs India Pvt. Ltd. — *September 2024 – June 2026*

- Built scalable backend services using **Spring Boot (Java)** and **FastAPI (Python)** for GenAI and transactional systems.
- Designed **REST and WebSocket APIs** supporting real-time, high-frequency workloads.
- Developed and integrated **LLM pipelines** (Mistral, LLaMA, Gemini, OpenAI) for automation and personalization.
- Optimized database queries and backend workflows, **reducing average API response latency by 40%**.

Software Development Intern — Bengaluru, India
Octopi Labs India Pvt. Ltd. — *March 2024 – August 2024*

- Developed backend services using **Spring Boot and Hibernate** for production applications.
- Integrated **DocuSign APIs** to enable multi-party digital agreement workflows.
- Designed and developed APIs for broker–tenant–institution interactions, improving process automation.

Intern — Pune, India
Centre for Development of Advanced Computing (C-DAC) — *July 2023 – August 2023*

- Built and tested **5G simulation environments** using Open5GS and UERANSIM.
- Executed end-to-end validation across multiple network slices, ensuring isolation and performance.
- Simulated real-world traffic to evaluate latency, reliability, and resource allocation.

KEY PROJECTS

DocuVerse | *Python, FastAPI, React, NVIDIA NIM, MySQL, Pinecone, Redis, AWS S3*

- Developed a **RAG-based full-stack application** with FastAPI and React, streaming responses in real time via SSE.
- Built multi-stage retrieval: LLaMA query expansion, **hybrid vector search**, and FlashRank reranking.
- Implemented a "small-to-big" chunking strategy combining Pinecone and MySQL for improved context relevance.
- Used Redis locks to handle concurrent sessions and prevent data conflicts.

aiRen Style | *Java, Spring Boot, Google Cloud Vision, Gemini APIs, Vector Search, MySQL, MongoDB, AWS S3*

- Built an **asynchronous GenAI pipeline** in Spring Boot for multi-stage AI fashion generation workflows.
- Integrated Google Cloud Vision and Gemini (text, multimodal, image) for face/body detection, parallelized content generation, and virtual try-on.
- Mapped generated outputs to real-world products using semantic search and Vision APIs.
- Designed dual storage: MySQL (transactions) and MongoDB (vector search), with AWS S3 for assets.

Marola.Life | *Java, Spring Boot, MySQL, Redis, AWS S3, Stripe, Wise, Firebase FCM*

- Built the backend for a digital content platform with a **multi-level commission structure** using Spring Boot.
- Designed a multi-level commission engine on MySQL; ensured atomicity of concurrent wallet transactions with **Redis distributed locking**.
- Integrated Stripe (subscriptions) and Wise (cross-border payouts) for financial workflows.
- Managed the digital content lifecycle (audio, video, blogs) through scalable storage pipelines.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, TypeScript

Backend: Spring Boot, FastAPI, Hibernate, REST APIs, WebSockets, Server-Sent Events (SSE)

AI/GenAI: LLM Integration, Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, Hybrid Search, Vector Search, Reranking

Databases: MySQL, MongoDB, Pinecone, Redis

Tools & Infrastructure: Docker, AWS S3, Git, Postman, Stripe, Firebase FCM

ACHIEVEMENTS & PUBLICATIONS

- **Patent published** (Application No. 202331068011): IoMT hospital network system with OpenFlow QoS and Cloud/Fog offloading, co-developed during MeitY-funded research.
- Contributed to the MeitY-funded project "Ensuring QoS in Internet of Medical Things using Software-Defined Networking." (Certificate)
- **GATE CS 2024** qualified (Scorecard).